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Ethnomedicinal plants used to cure various diseases by the local people of Balasore district of Odisha

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Abstract-The medicinal plant plays an important role for curing various diseases of common people specially in the tribals and rural communities of Odisha. Ethno-medicinal research carried out to collect, identify and document information on the medicinal plants traditionally used to treat various diseases by the local inhabitants of Balasore district, Odisha, India. Different plants that expected to have medicinal value, were collected and identified by botanical classification. 40 Plant species belonging to 33 different families were recorded in the interviews and enquiries. The plant parts that were more commonly utilized for the preparation of ethno-medicine were the leaves (39.68%), roots (14.28%), bark (12.69%), whole plants (9.52%) and fruits (7.93%).

Key words: Ethno-medicinal, Balasore district, tribal communities, document etc.

INTRODUCTION

'Ethnobotany' generally refers how a people of particular region and culture makes the use of indigenous plants for various purpose including medicine. In the growing society, there is often occurrence of various new disease and treatment of such disease through allopathic medicine is very expensive and have side effects. Medicinal plants are traditional and hence easily accessible and affordable source of primary health care specially for the rural and tribal people, who cannot afford or access basic healthcare. Rural people are far away from the modern medical facilities and that's why they use many locally available plants species for the treatment of various diseases.

The State of Odisha located between the parallels of 17° 49'N and 22° 34'N latitudes and meridians of 81° 27'E and 87° 29'E longitudes, surrounded by the adjacent states of West Bengal to the north-east, Jharkhand to the north, Chhattisgarh to the west and north-west, Telangana to the south-west and Andhra Pradesh to the south.[1] This paper aimed at collecting the data on ethnomedicinal plants of Balasore district, their traditional uses and documentation of such information for further use.

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MATERIALS & METHODS

STUDY AREA

Balasore is one of the coastal districts of Odisha located at 20° 48' to 21° 59' North Latitude and 86° 16' to 87° 29' east Longitude.[2] The district covers an area of 3634 sq. kms having total population of 23,17,419 as per 2011 census. The district is surrounded by Medinipur district of West Bengal in its northern side, Bay of Bengal in its east, Bhadrak district in its south and Mayurbhanj and Kendujhar district lies on its western side.

DATA COLLECTION

The field study was carried out from November 2022 to November, 2023 in various villages of Balasore district including major areas such as Baliapal, Basta, Bhograi, Soro, Nilagiri, Khaira etc. The data were collected from the local Vaidya, ayurvedic practitioner and old knowledgeable person from various villages as they have been using this traditional medicine since a long time. Informal conversation and personal interviews were done with around 75 people out of which 64% were male and 36% were female.

Several information such as different kind of plants they used to cure various diseases, preparation, mode of administration, doses, location of plant species, plant parts used etc. were recorded from the villagers.[3] The plant families, genera and species of the collected plant specimen were identified using standard floras, available literature[4] and with the help of professor of DSPM University. The specimen of the identified species was deposited in the herbarium of Botany Department, DSPM University, Ranchi, Jharkhand. The list of medicinal plants were arranged in tabular form along with their botanical names followed by family & life form, local names in Odia, the plant parts used and their way of intake to cure various diseases.[5]



Fig. 1- Map of Odisha (Red colored part indicating Balasore)



Fig. 2- Map of Balasore district

RESULT & DISCUSSION

The village people and tribes of Balasore district, diagnose the diseases by their experience which is quite good because they live in a remote area that lacks any modern medical facilities and treat the disease using medicinal plant. Documentation of such plants is quite important for the preservation of plants and indigenous knowledge system.[6] A total of 40 plant species belonging to 33 different families were found which are used as medicine. The most cited medicinal plant families were Euphorbiaceae, Amaranthaceae, Rutaceae, Verbnaceae, Lauraceae, Liliaceae etc. The reported plant species were used to treat 56 different disease/health conditions. The most often cited health conditions were: Cold, fever, asthma, inflammation, wound healing, indigestion, diarrhea, diabetes, rheumatism, piles etc.[7] Moreover, out of 40 plant species, 8 plant species were used for wound healing & indigestion, 5 plant species for cold, 7 plant species for fever, 4 plant species for inflammation, asthma & diabetes, 3 plant species for diarrhea & piles and several plant species for other kind of diseases. Among the medicinal plants, herbs (42.5%) were most frequently used followed by trees (30%), shrubs (15%) and climber (12.5%) which is represented in figure 3. Most of the plant parts are consumed orally after processing such as crushing into juice, or grinding into powders or mixing the extracts with honey.[8]

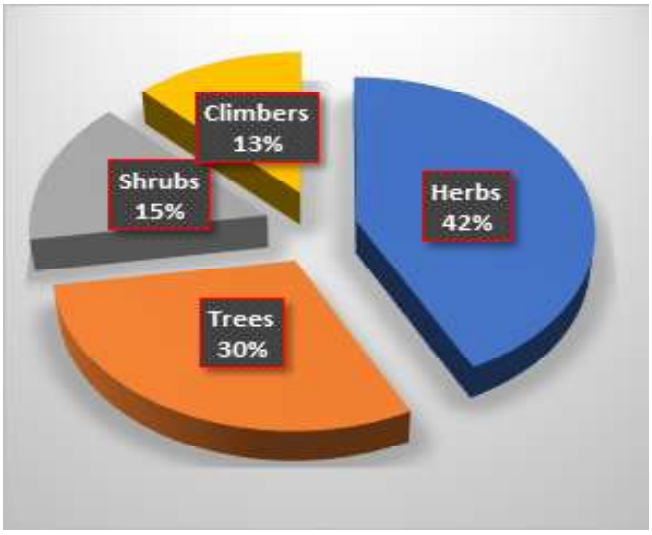


Fig. 3- Growth form of recorded plant species

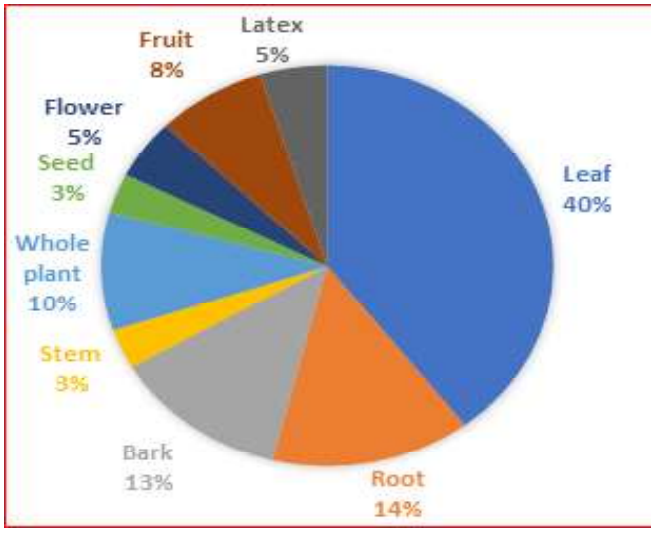


Fig. 4- Plant parts used in preparing medicines

Table 1- Ethno-medicinal plants of Balasore district, Odisha

Sl. No.	BOTANICAL NAME	FAMILY & LIFE FORM	LOCAL NAME	PLANT PARTS USED & THEIR MEDICINAL VALUE
01	*Acalypha indica* L.	Euphorbiaceae Herb	Indramarisha ଇନ୍ଦ୍ରମାରିଷ	The leaf juice is used as nasal drop in headache. Leaf powder is used for maggot-infested wounds. A mixture of the leaves and castor oil is applied to relieve joint aches.
02	*Achyranthes aspera* L.	Amaranthaceae Herb	Apamarga ଅପମାର୍ଗ	Leaf juice is applied to heal the fresh wounds. Decoction of seed is used as an antidote against dog, jackal and cat bite. Leaf juice with a pinch of salt is applied to cure skin infection due to ring worm.[9]
03	*Albizzia lebbeck* Benth.	Mimosaceae Tree	Sirisa ଶିରିଶ	2 gm bark powder mixed with 5ml of bark decoction of Arjuna (*Terminalia arjuna*) and 5 drops of honey is given once a day for one month to cure mental retardation and epilepsy.
04	*Allium sativum* L.	Amaryllidaceae Herb	Rasuna ରସୁଣ	Crushed garlic mixed with hot mustard oil is applied to the lower foot to cure the common cold. Crushed garlic with a pinch of salt has antimicrobial activity and can reduce toothaches.
05	*Aloe vera* (L.) Burm. f.	Asphodelaceae Herb	Aloe Vera ଆଲୋ ଭେରା	The crushed fresh leaves are used to treat burns and wounds. The gel obtained from leaves is used to improve skin conditions and also taken orally to improve digestive health.
06	*Alternanthera philoxeroides* (Mart.) Griseb.	Amaranthaceae Herb	Madaranga ମଦରଙ୍ଗା	Decoction (100 ml) of the whole plant with one cup of milk once a day for 15 days is prescribed to feeding mother to increase lactation. Juice of stems and leaves is used to cure eye disease.
07	*Argemone mexicana* L.	Papaveraceae Herb	Kanta agara କଣ୍ଟା ଅଗର	The latex of the stem is applied externally to cure wounds of leprosy.[10]
08	*Asparagus racemosus* Willd.	Liliaceae Climber	Satabari ସତାବରୀ	Tuberous root powder (5g) mixed with 5 drops of honey is given to improve health and restore potency.
09	*Azadirachta indica* A. Juss.	Meliaceae Tree	Nimba ନିମ୍ବ	Fresh leaves are mixed with hot bathing water to cure various skin diseases in body. The bark of the neem tree has antiseptic and astringent properties, which helps heal wounds.[11]
10	*Bacopa monnieri* (L.) Wettst.	Plantaginaceae Herb	Brahmi ବ୍ରାହ୍ମୀ	Whole plant is crushed to make juice and taken orally to increase memory power. The leaf powder is used to reduce hairfall.
11	*Calotropis gigantea* (L.) W.T. Aiton	Asclepiadaceae Herb	Arakha ଆରଖ	3-4 mildly warmed leaf with castor oil are bandaged over testicles everyday against hydrocele. The milky latex is applied to treat wounds caused by spines.
12	*Centella asiatica* (L.) Urb.	Apiaceae Herb	Thalkudi ଥାଲକୁଡି	The whole plant is widely used as a blood purifier as well as for treating high blood pressure, for memory enhancement and promoting longevity.
13	*Cheilocostus speciosus* (J. Koenig) C.D. Specht	Costaceae Herb	Chakrakedara ଚକ୍ରକେଦାର	The rhizome is used to treat fever, rash, asthma, bronchitis in Ayurveda. The leaves are used to control diabetes.
14	*Cinnamomum Zeylanicum* Blume	Lauraceae Tree	Dalchini ଡାଳଚିନି	The bark is used for gastrointestinal upset, diarrhea, and gas. It is also used for stimulating appetite; for infections caused by bacteria and parasitic worms; and for menstrual cramps, the common cold.

15	*Cissus quadrangularis* L.	Vitaceae Climber	Hadabhanga ହାଡଭଙ୍ଗା	The paste of Whole plant is applied over bone fractured part once daily till cure. Leaf paste is taken orally to cure stomach-ache.[12]
16	*Curcuma longa* L.	Zingiberaceae Herb	Haladi ହଳଦୀ	The paste of rhizome is applied on the face for clear and glowing skin. The rhizome extract has antimicrobial activity and used as anti-inflammatory agent.
17	*Cynodon dactylon* (L.) Pers.	Poaceae Herb	Duba ଦୁବ	The leaf juice is taken to cure blood vomiting. Paste of the whole plant is applied on chronic wounds.
18	*Euphorbia hirta* L.	Euphorbiaceae Herb	Dudhi ghasa ଦୁଧୀ ଘାସ	The decoction of flowers and fruits are used for asthma and respiratory tract infections. The latex is used for healing of wounds.
19	*Feronia elephantum* Correa	Rutaceae Tree	Kaitha କଇଥ	Matured fruits are taken for 10-15 days for digestion.[13]
20	*Ficus religiosa* L.	Moraceae Tree	Aswastha ଅଶ୍ୱତ୍ଥ	Bark paste is applied to control rheumatic joint pain and tetanus.
21	*Gloriosa superba* L.	Liliaceae Climber	Agnisikha ଅଗ୍ନିଶିଖା	Root is made into a paste with fresh cow urine and applied on piles twice a day for 7 days. Leaf powder is used for skin diseases.
22	*Heliotropium indicum* L.	Boraginaceae Herb	Hati sundha ହାତୀ ଶୁଣ୍ଢ	The leaf juice is used to treat the stings & boils of scorpion & insect bites. The boiled juice with castor oil is used to treat mad dog bite infections.
23	*Lantana camara* L.	Verbanaceae Shrub	Dahanimari ଡାହାଣିମାରି	Leaf juice is applied externally to treat cut and wounds. Leaves are boiled and applied for swellings. Bark is an astringent and used as a lotion for leprous ulcers.
24	*Litsea glutinosa* (Lour.) C.B. Rob.	Lauraceae Tree	Pimpala ପିମ୍ପଳା	The leaf extract is taken orally to treat diarrhea, abdominal pain, indigestion etc. The root, bark and leaves are used medicinally to reduce fever, reduce swelling, and treat diarrhea.[14]
25	*Millettia pinnata* (L.) Panigrahi	Fabaceae Tree	Karanja କରଞ୍ଜ	Dried flower powder is taken orally to reduce blood sugar. Juice extracted from green fruits is mixed with mustard oil and applied in case of rheumatic pain and fresh bark extract is administered orally to cure bleeding piles. Seeds are grinded with black pepper and taken to reduce fever. Two spoonful of leaf decoction is taken orally to cure asthma.
26	*Momordica charantia* L.	Cucurbitaceae Climber	Kalara କଲରା	The paste of seven leaves is given daily on empty stomach for diabetes.
27	*Moringa oleifera* Lam.	Moringaceae Tree	Sajana ସଜନା	Leaves are boiled in water and one tumbler is taken early in the morning in empty stomach for hypertension.
28	*Murraya koenigii* (L.) Spreng.	Rutaceae Tree	Bhrusanga ଭୃସଙ୍ଗ	The green leaves of are used in treating piles, inflammation, itching, fresh cuts, dysentery, bruises, and edema. The roots are purgative to some extent.
29	*Nyctanthes arbortristis* L.	Nyctaginaceae Tree	Gangasiuli ଗଙ୍ଗଶିଉଳି	Fresh leaf juice mixed with honey is given to cure malarial fever.
30	*Ocimum sanctum* L.	Lamiaceae Herb	Tulasi ତୁଳସୀ	The leaf of juice useful against constipation, cholera and leprosy. Leaf taken with honey is useful against common cold.
31	*Opuntia dilleni* (Ker Gawl.) Haw.	Cactaceae Shrub	Nagapheni ନାଗଫେଣୀ	Decoction of the pulp is applied externally for eye disease.
32	*Phyllanthus niruri* L.	Phyllanthaceae Herb	Bhuin-Anla ଭୂଇଁ ଅଁଳା	The whole plant extract is used to cure liver diseases and to dissolve kidney stones.
33	*Piper betle* L.	Piperaceae Climber	Pana ପାନ	The juice of the leaves is taken orally to cure fever, common cold & to improve digestion.
34	*Plumbago zeylanica* L.	Plumbaginaceae Shrub	Raktachitrak ରକ୍ତ ଚିତ୍ରକ	Two teaspoonful root powder is taken after meal for indigestion.[15]
35	*Rauvolfia serpentina* (L.) Benth. ex kurz	Apocyanaceae Shrub	Patalagaruda ପାତାଳଗରୁଡ଼	One teaspoonful root powder mixed with black pepper is taken with a cup of water twice a day for two days to cure snake-bite. Leaf Juice mixed with the juice of *Andrographis paniculata, Azadirachta indica* and honey is given for 7 days to cure malaria.
36	*Ricinus communis* L.	Euphorbiaceae Tree	Jada ଜଡ଼	The oil prepared from the fruit greatly helps in the regrowth of hair. Castor oil is also applied in case of abdominal pain.
37	*Terminalia chebula* Retz.	Combretaceae Tree	Harida ହରିଡ଼	One teaspoonful of fruit powder is given with warm water once daily before going to bed to increase appetite and good digestive health.
38	*Tridax procumbens* L.	Asteraceae Herb	Bisalyakarani ବିଶଲ୍ୟକରଣୀ	The juice extracted from the leaves is directly applied on wounds to heal. The leaf powder is taken orally to treat diabetes.[16]
39	*Vitex negundo* L.	Verbenaceae Shrub	Begunia ବେଗୁନିଆ	Tender leaves juice is given for cold and fever. Powdered flowers are given with milk to treat diarrhea and cardiac disorders.
40	*Withania somnifera* (L.) Dunal	Solanaceae Shrub	Aswagandha ଅଶ୍ୱଗନ୍ଧା	The root powder boiled with cow milk is used for adenopathy, asthma and hypertension. The leaves are used to cure tuberculosis.

CONCLUSION

In conclusion, the ethnobotanical study of plants used to treat various diseases provides valuable insights into traditional medicinal practices. Documenting the knowledge and practices of local communities, this study helps in the sustainable utilization of plant resources and development of plant based medicinal products for better health care of Balasore as well as Odisha.

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